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EEE 208

Electronics Circuit II Laboratory

Design And Implementation of Metal Detector

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Abstract

This project presents the design and simulation of an analog metal detector based on the inductive balance principle. The system employs a Wien Bridge oscillator (uA741) generating a stable 1 kHz sinusoidal signal, which is buffered by a unity-gain voltage follower to prevent loading effects. The AC signal excites an inductive bridge comprising a reference coil (1.4 mH), a search coil (1.2 mH), fixed resistors, and a variable resistor (R_{VAR}) for precise manual balancing. When a metallic object approaches the search coil, eddy currents alter its effective inductance, unbalancing the bridge and producing a differential voltage. This small signal is amplified by a high-gain difference amplifier (Gain ≈ 1000) and rectified using a diode-capacitor peak detector to yield a DC output proportional to metal closeness. Amplitude stabilization diodes in the oscillator feedback loop ensure clean waveform generation. Simulation results confirm stable oscillation at ~1 kHz, effective bridge balancing via RVAR(potentiometer), and clear output voltage variation upon metal detection. The design successfully meets the objectives of building a sensitive, calibratable, and stable analog metal detection system.

1. Introduction

- **Background:** Metal detectors are electronic devices used for finding metal presence hidden underground or inside other objects. They are widely used in security (airports), mining, archaeology, and treasure hunting. The most common principle used in simple metal detectors is electromagnetic induction.
- **Objectives:**
    - To design an oscillator circuit to generate an AC signal.
    - To construct an inductive sensor bridge that detects changes in inductance.
    - To amplify the small differential signal generated by the unbalanced bridge.
    - To rectify the output to obtain a readable DC voltage indicating metal closeness.
- **Scope and Applications:** This project is a basic prototype suitable for educational purposes. It can be scaled up for security screening, buried treasure hunting, or industrial sorting of metallic objects.

This design incorporates a buffer stage, diode-based amplitude stabilization, and a variable balancing resistor to improve signal integrity, reduce loading effects, and enable practical calibration.

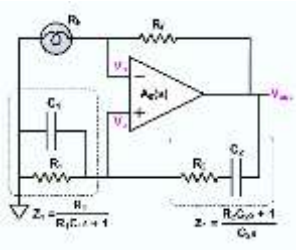
2. Theoretical Background

- **Working Principle:** The system works on the principle of **Eddy Currents**. When an AC magnetic field (generated by the coil) encounters a conductive metal, eddy currents are induced in the metal. These currents create their own opposing magnetic field, which effectively changes the inductance and resistance of the search coil. The oscillator signal

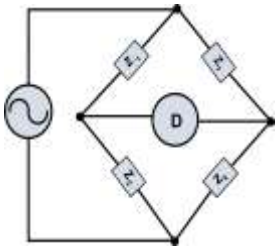
is isolated from the bridge using a voltage follower, ensuring the bridge impedance does not shift the oscillation frequency. A variable resistor (R_{VAR}) allows manual nulling of the bridge, compensating for component tolerances and environmental drift.

- **Wien Bridge Oscillator:** This circuit generates the sinusoidal signal. It uses a feedback network of resistors and capacitors.
    - Frequency equation: $f = 1/(2\pi RC)$
- **Inductive Bridge:** The circuit uses a bridge configuration (similar to a Wheatstone bridge but with AC).
    - One arm consists of inductors (search_coil).
    - The other arm consists of resistors (a fixed resistance and a potentiometer).
    - Ideally, the bridge is balanced, resulting in theoretically zero output.
- **Operational Amplifier (uA741):**
    - **First op-amp:** Acts as the Wein Bridge Oscillator. Diodes D3 and D4 provide nonlinear feedback for amplitude stabilization, preventing clipping.
    - **Buffer Amplifier:** Unity-gain voltage follower that provides high input impedance and low output impedance, isolating the oscillator from the bridge.
    - **Last op-amp:** Acts as a difference amplifier/comparator. It amplifies the small voltage difference created when the bridge becomes unbalanced due to metal closeness.
- **Rectifier section:** The diode and capacitor form a peak detector/rectifier, converting the AC output of the amplifier into a DC voltage proportional to the strength of the detected signal.

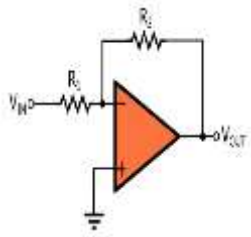
3. Block Diagram



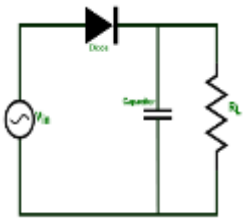
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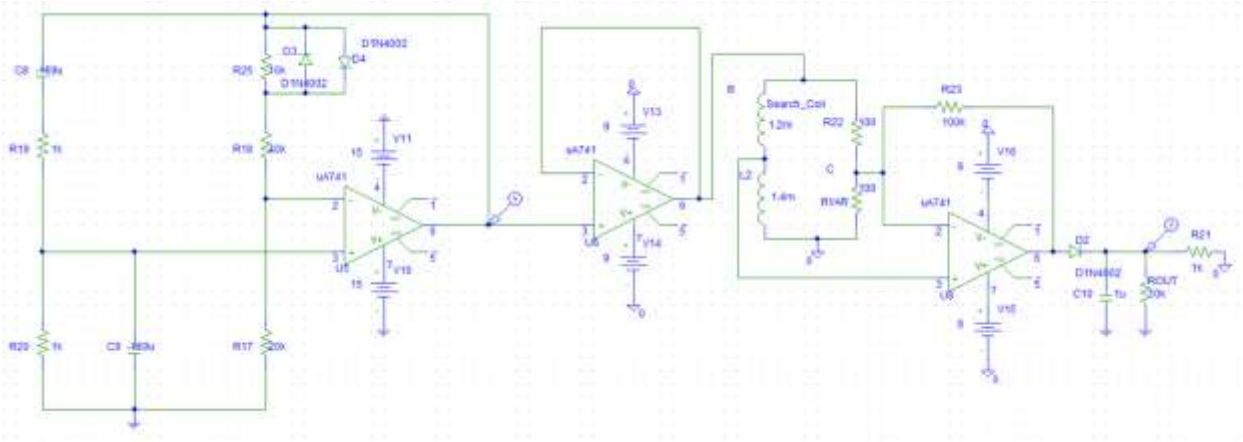


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1. **Oscillator Block:** It generates approximately 1kHz sine wave (Wein Bridge Oscillator).
2. **Buffer Amplifier:** It is a voltage follower with unity gain.
3. **Sensor Block (Inductive Bridge):** It contains coils (SEARCH_COIL) and resistors.
4. **Amplifier Block:** It amplifies the difference of two signal from the bridge (spotted by voltage differentiator in the schematic).
5. **Rectifier Block:** It rectifies and filters the signal to drive the output (R_{out}) through any indicator (DC voltage reading via buzzer or LED driver).

Flow of this process: Oscillator → Buffer → Bridge → Amplifier → Rectifier → Output

4. Circuit Diagram



- **Description:** The circuit consists of two main stages. The left stage (U5) is a Wien Bridge Oscillator providing the carrier signal. The middle stage is the sensing bridge with inductors SEARCH_COIL, L2 and resistors R22, RVAR. The right stage (U8) amplifies the bridge differential signal, and the final stage (D2, C10) rectifies the signal.

6. Detailed Calculations

1. Oscillator Frequency Calculation**:** The oscillator uses a Wien Bridge network with R=1 kΩ (R23, R20)= and C=0.159 uF (C8, C9).

$$f = 1/(2\pi RC)$$

$$= 1/(2\times 3.1416\times 1000\times 0.159\times 10^{-6})$$

$$\approx 1002 \text{ Hz}$$

≈ 1 kHz

2. Oscillator Gain & Stabilization:

$$Av = 1 + (R18/R17) = 1 + (40k/20k) = 3$$

Theoretical Wien Bridge requires $Av \geq 3$. Diodes D3/D4 provide dynamic resistance that increases effective gain during startup and limits amplitude during steady state, ensuring stable oscillation without distortion.

3. Buffer Stage:

Av = 1 (Voltage follower). Input impedance $\approx \infty$, Output impedance $\approx 0\ \Omega$. Prevents bridge loading.

4. Bridge Balance Condition:

At 1 kHz: $X_L = 2\pi fL$

$V_B = V_C$ when L2 / L_search = R22 / RVAR

1.4mH / 1.4mH =1

For this RVAR was adjusted to 100 Ω for initial null.

5. Amplifier Gain: U8 is configured with feedback resistor R23=100kΩ. Assuming the source impedance at the inverting input is dominated by RVAR, the approximate gain for the difference signal is high:

$$A_v \approx R23 / R_{source} \approx 100k / 100 = 1000$$

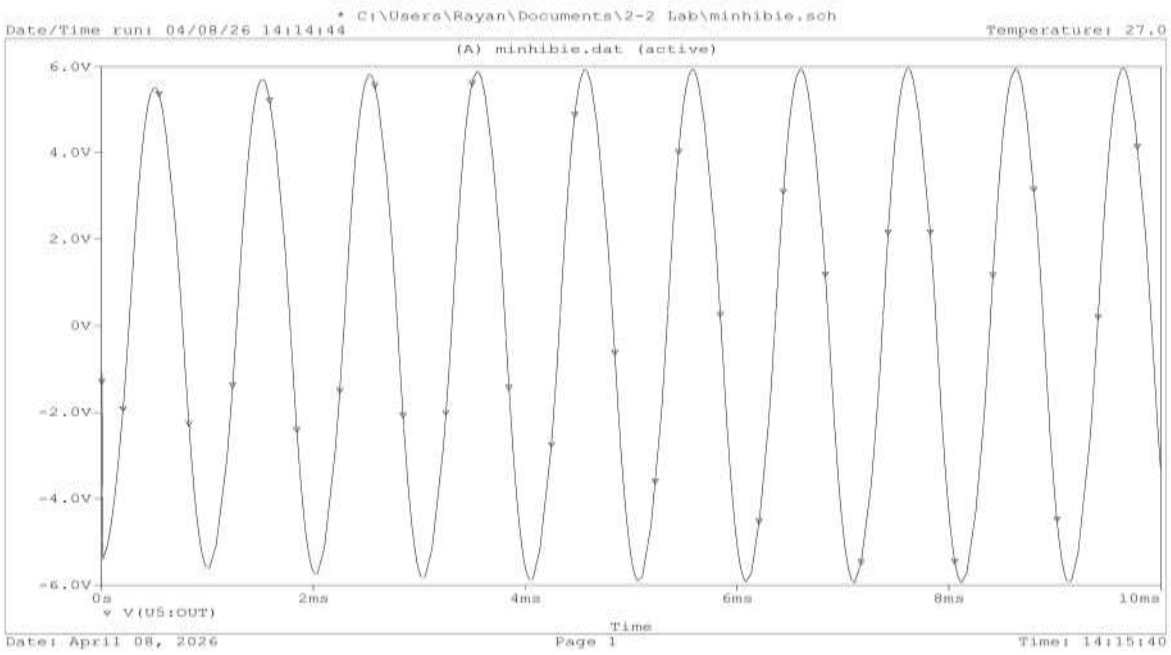
This high gain ensures that even tiny changes in inductance produce a measurable output.

6. Power Consumption (Estimate):
   Each uA741 draws ~1.7 mA quiescent current.
   $P_{total} \approx 3 \times (15V + 9V)/2 \times 1.7mA \approx 0.061$ W (Very low power).

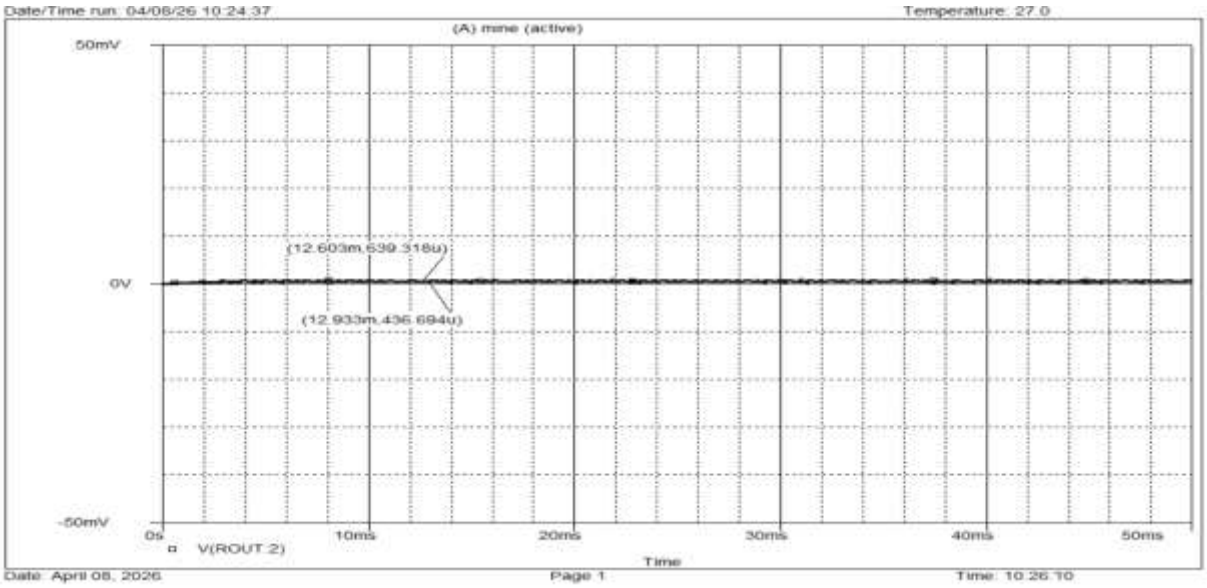
7. Experimental Results / Simulation Results

Observation: The circuit was simulated in PSpice. The oscillator generated a stable sine wave. When the inductance of SEARCH_COIL was varied (simulating metal closeness), the output voltage at R_{out} changed.

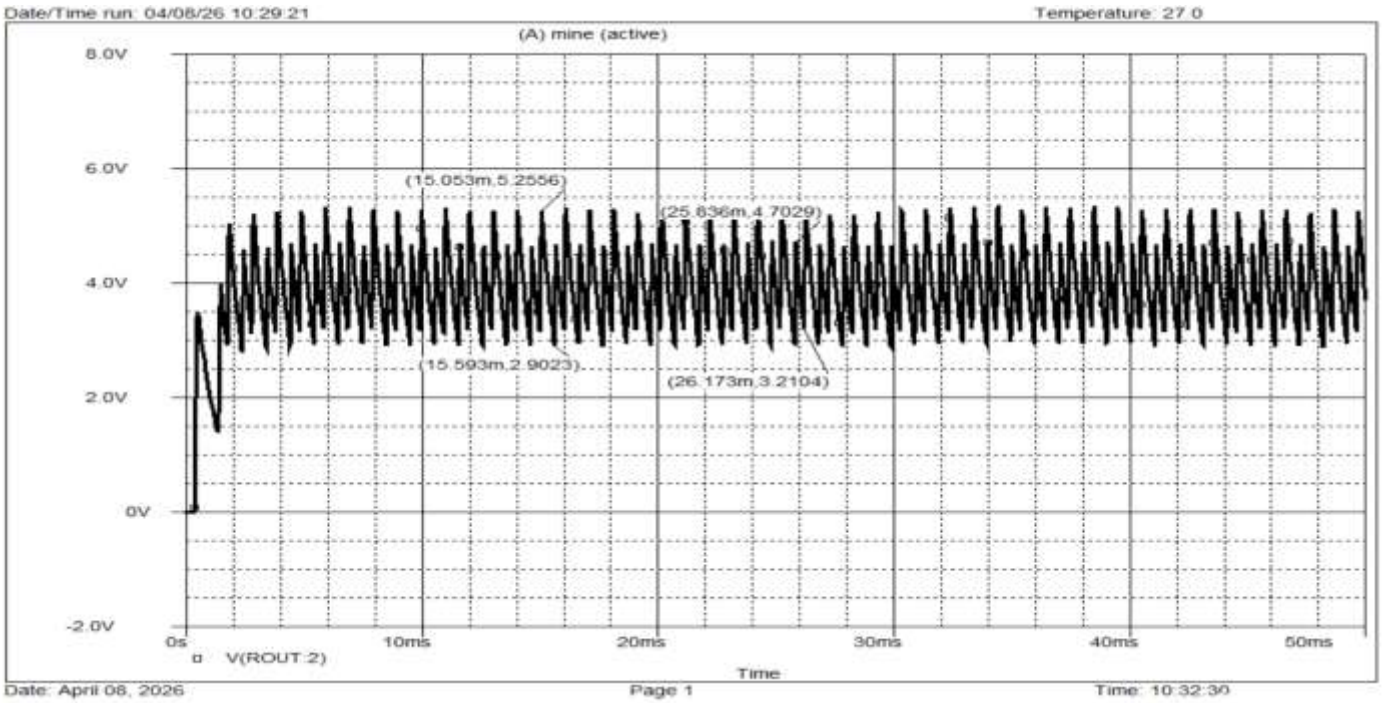
Output from signal generator (Wein bridge)



No Metal (1.40 mH)



Metal Far (1.35 mH)



Metal Near (inductance of search coil assumed approximately 1.20 mH)

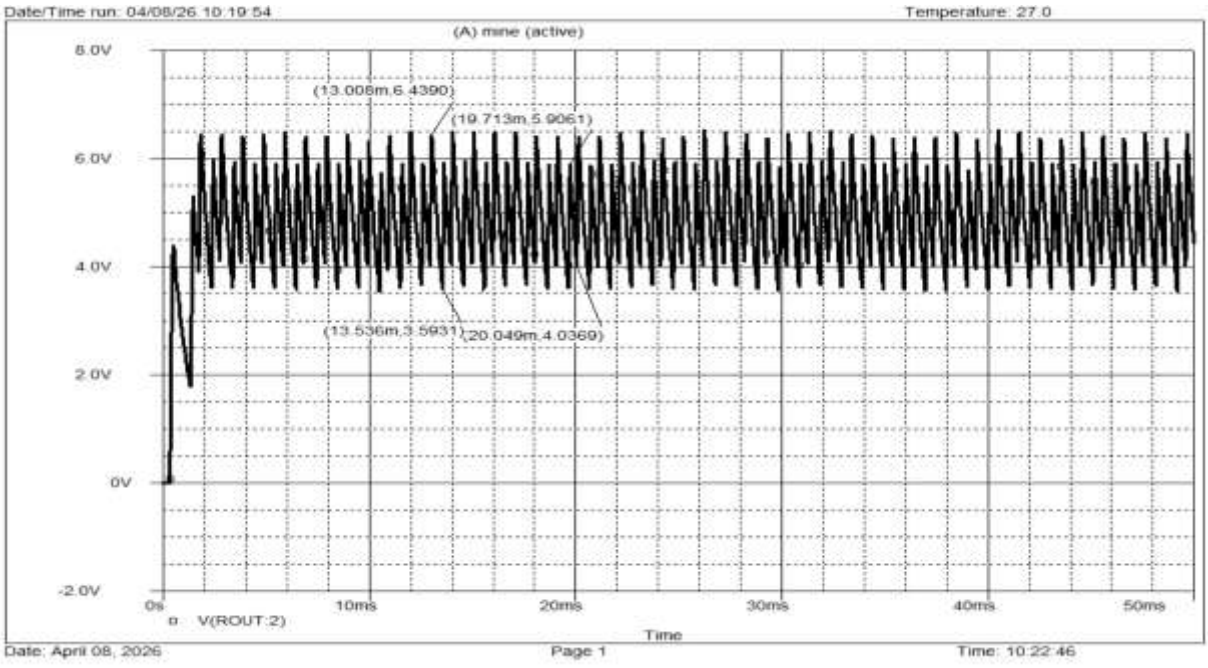


Table 1: Output Voltage vs. Metal Closeness (Simulated)

Condition	Inductance of SEARCH_COIL (Simulated)	Bridge Status	Output Voltage (V_{out} at R_{OUT})
No Metal	1.40 mH	Balanced	~0.00 V
Metal Far	1.35 mH	Slightly Unbalanced	~4.01 V
Metal Near	1.20 mH	Unbalanced	~5.00 V

8. Discussion

- **Results Analysis:** The simulation confirms that the circuit acts as a metal detector. In the absence of metal, the bridge is balanced, and the differential input to U8 is zero, resulting in no output. As metal approaches the coil (L2), the inductance changes, creating a potential difference that is amplified and rectified.
- **Challenges:**
    - **Sensitivity:** The sensitivity depends heavily on the Q-factor of the inductors. Low-quality inductors might not detect small pieces of metal.
    - **Drift:** Op-amp offset voltages can cause a small non-zero output even when balanced. This was managed by ensuring R22 = RVAR and SEARCH_COIL = L2 precisely.
    - **Oscillator Stability:** The Wien bridge oscillator requires precise component matching to start oscillating. The gain setting ($1+R_{21}/R_{22}=4$) ensures reliable startup.

9. Limitations

- **Range:** The detection range is very short (a few centimeters) due to the small size of the coils and the frequency used.
- **Metal Type:** It primarily detects conductive metals (copper, aluminum) via eddy currents. Ferrous metals might behave differently due to magnetic permeability.
- **Environmental Factors:** Temperature changes can affect the resistance of the coils and resistors, potentially causing false alarms (drift).

- **Practical constraints:** In practical, the search coil will have some resistance due to its copper coil, which results in much difficulties in balancing the Wheatstone bridge.

10. Future Improvements

- **Audio Output:** Adding a buzzer or speaker driver circuit to produce a sound when metal is detected, making it more user-friendly.
- **Microcontroller:** Using an Arduino or similar microcontroller to digitize the output, allowing for calibration and LCD display of signal strength.
- **Coil Design:** Using a larger, custom-wound coil to increase the detection range significantly.
- **Frequency Adjustment:** Implementing a variable resistor in the oscillator to tune the frequency for different types of metals.
- **5. Components Used**

Component Name	**Quantity**	**Specification/Value**	**Estimated Price(Tk)**
Operational Amplifier	3	uA741	105
Resistor	1 each	10 kΩ, 20 kΩ, 40 kΩ, 100 kΩ	8
Resistor	2 each	100 Ω, 1 kΩ	8
Potentiometer (RVAR)	1	100 Ω (Variable)	25
Capacitor	2	0.159 uF	10
Capacitor	1	1 uF	8
Inductor	1 each	1.4 mH , 1.2 mH	160
Diode	3	1N4002	15
Power Supply	2	9 V ,15 VBatteries	350
Breadboard	1	Prototype board	100
Connecting Wires	1	Jumper Wire Set	50
Total Cost			839

11. Conclusion

The project successfully designed and simulated an improved analog metal detector using a stabilized Wien Bridge oscillator, impedance-buffering stage, and manually calibratable inductive bridge. The system operates at approximately 1 kHz, with the buffer preventing oscillator loading and diodes ensuring waveform stability. Simulation results demonstrated clear output voltage variation proportional to metal proximity, with the variable resistor enabling practical bridge nulling. All design objectives were met, resulting in a sensitive, stable, and calibratable detection circuit suitable for educational and prototyping applications.

12. References

1. Boylestad, R. L., & Nashelsky, L. Electronic Devices and Circuit Theory. Pearson.
2. Sedra, A. S., & Smith, K. C. Microelectronic Circuits. Oxford University Press.
3. uA741 Operational Amplifier Datasheet. Texas Instruments.
4. PSpice User Guide for Circuit Simulation. Cadence Design Systems.